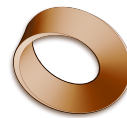


Mapping Your Journey

Forces 1 — Kinematics • §6.5.4.1 • AQA Trilogy 8464

Distance, displacement, and the vector that connects them



You are going to measure a journey you already know — the one from your home to school.

You will get **two different numbers** for that single journey, from **two different tools**. The whole of this topic lives in the gap between them.

You need: a laptop, your school Google account, and your exercise book.

Predict first

Before you open anything, write in your book: which will be bigger — the **distance you actually drive** to school, or the **straight line** from home to school?

By roughly how much? Twice? Ten times?

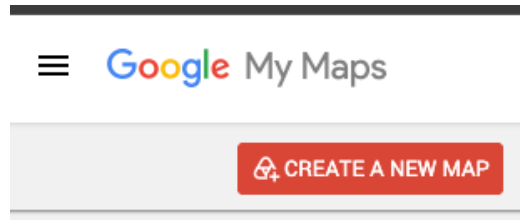
The credit is for the *reason*, not the number. Commit to it in writing now, before the map tells you the answer.

Part 1 — start your map

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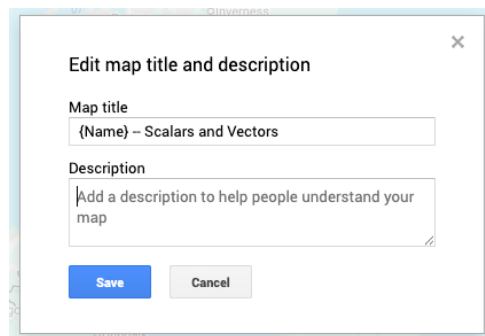
1 Create a new map

Go to `mymaps.google.com` and sign in with your **school Google account**. Click the red **Create a New Map** button.



2 Put your name on it

Click **Untitled Map** at the top left. Title it with **your own name** and *Scalars and Vectors*, then **Save**. Naming it after yourself is how you find it again next lesson.



Why two tools?

Google Earth can measure a straight line, but it will not give you **driving directions**. My Maps will — but it will not give you a **bearing**.

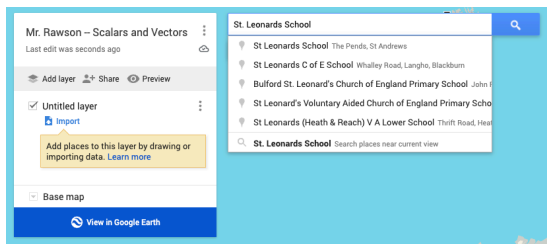
So you start here and finish in Google Earth. That is not an inconvenience; it is the lesson. The two tools answer two different questions about one journey.

Part 1 — add your two places

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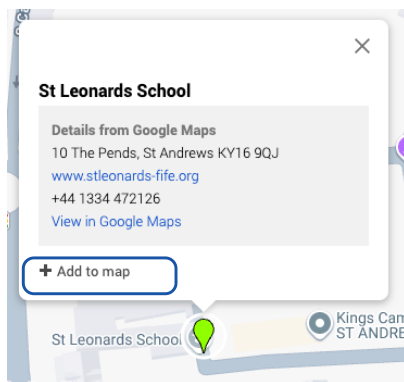
3 Search for the school

Type **St Leonards School** into the search bar. Take the one on **The Pends, St Andrews** — there are several other St Leonards schools in the list.



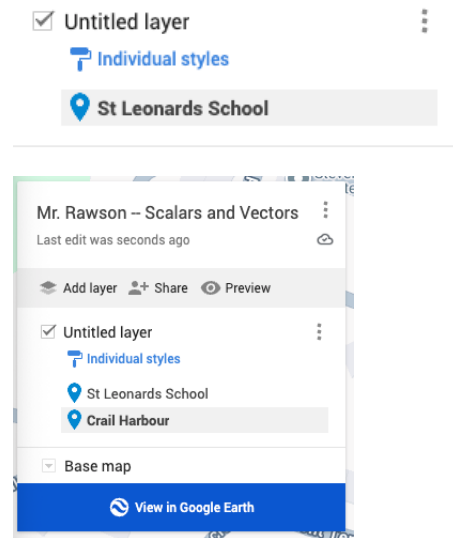
4 Click Add to map

The information card opens. **Add to map** sits at the bottom, ringed below. Miss it and the pin disappears the moment you search for anything else.



5 Now add home

It appears in the layer on the left. Repeat steps 3–4 for **your home**, and finish with **two pins**.

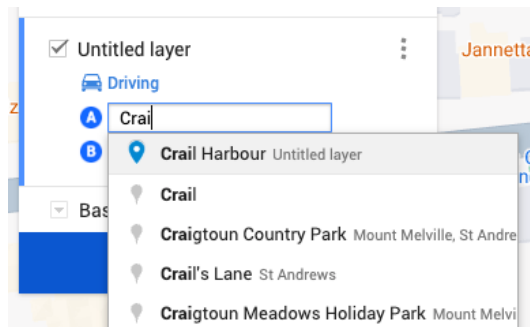


Part 1 — ask for driving directions

Mapping Your Journey

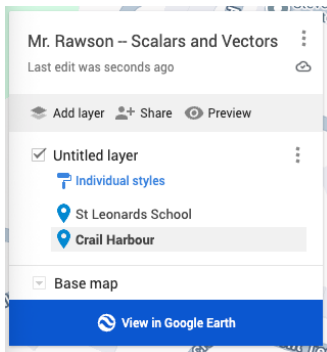
6 Directions to here

Click your **St Leonards School** pin, then the directions icon  on its card, and choose **Directions to here**. A new layer opens with an **A** box and a **B** box.



7 Type your home into A

Start typing and your own saved pin appears at the top of the list — take that one, not a similar name below it. A blue driving route is drawn between the two.

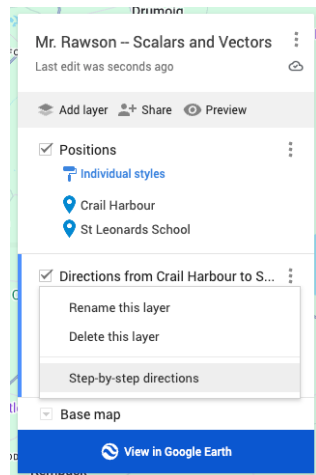


Look at the route

It follows **roads**. It bends, it doubles back, and it goes the long way round the coast wherever the road does.

That is exactly what you want. This is the journey you *actually* make — not the journey a crow would make.

Part 1 — read off your two numbers



8 Open the step-by-step directions

Click the **three dots** beside the *Directions from ...* layer, then **Step-by-step directions**. The route opens turn by turn, with the totals on the first line.

Write both of these in your book

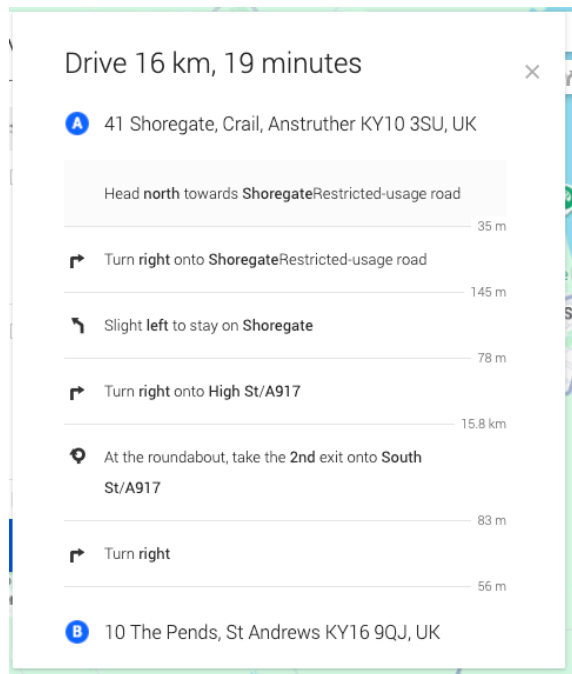
- ▶ How many **kilometres** is the drive to your house?
- ▶ How many **minutes** would it take to drive?

The kilometres are your **distance** — how far you really travel. Keep the minutes as well; you will want them when we get to speed.

9 Cross to Google Earth



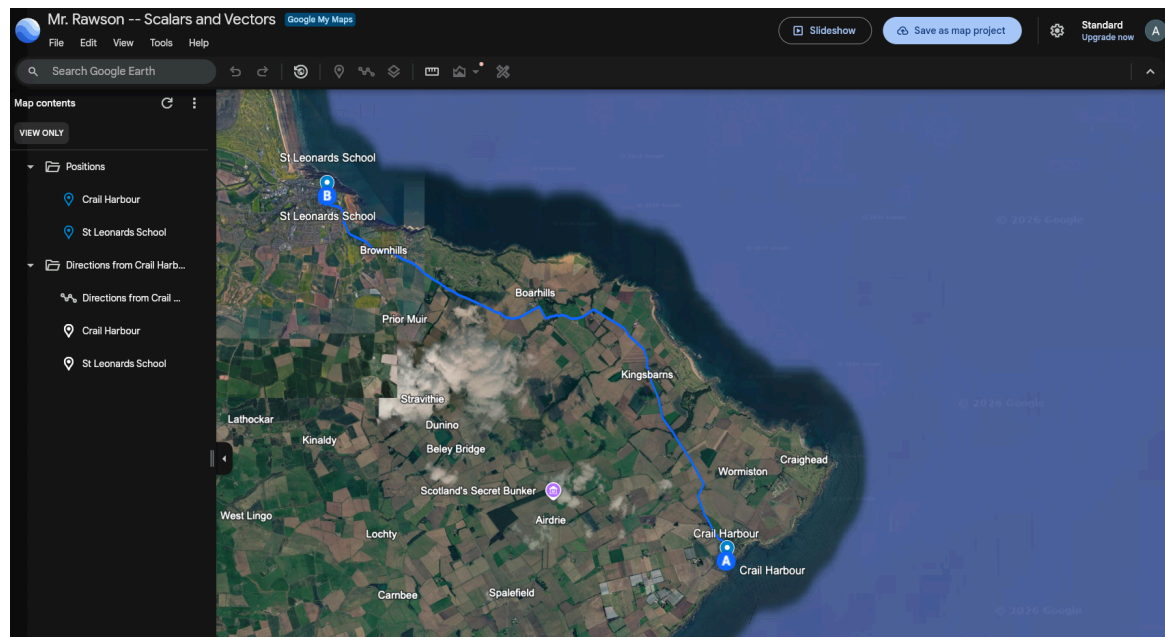
The button is at the bottom of the left-hand panel. Both your pins travel across with you.



*Mr Rawson's route: **16 km, 19 minutes**, and every turn in between.*

Part 2 — this is what you should be looking at

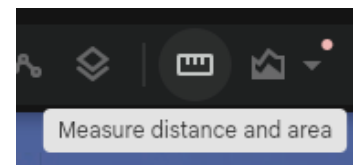
Mapping Your Journey



Everything travelled across with you: **both pins** under *Positions*, and the **blue driving route** you asked My Maps for.

Notice what the route does — it swings inland through Boarhills and Kingsbarns rather than following the coast, because that is where the road goes.

Find the Measure tool



It is the ruler icon in the top toolbar. Hover to check — the tooltip reads *Measure distance and area*.

Part 2 — draw the straight line

Mapping Your Journey

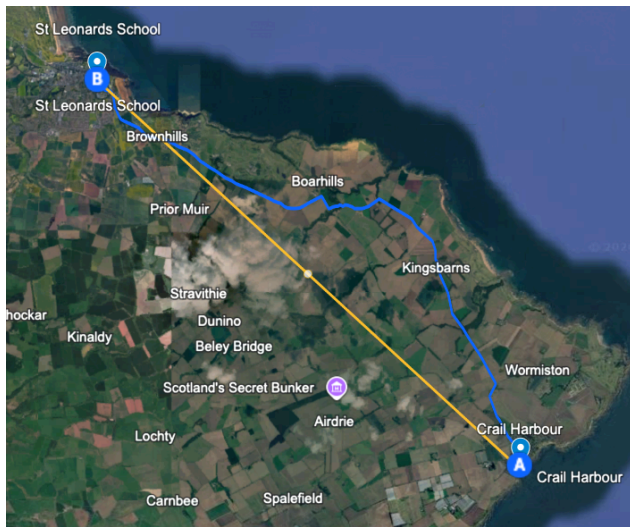
10 Draw it, A to B

With the Measure tool open, click once on the big **A** — your home. Then **double-click** the big **B** — school.

The double-click matters. A single click tells Google Earth you want to keep adding more line segments, and it will wait for the next one instead of giving you an answer.

You now have two lines

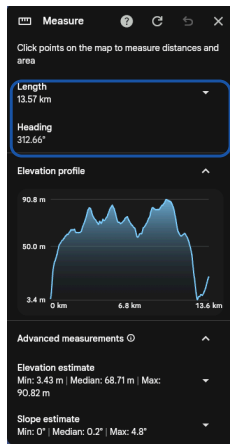
The **blue** one is the driving route. The **yellow** one is the straight line. Same two endpoints, two completely different paths.



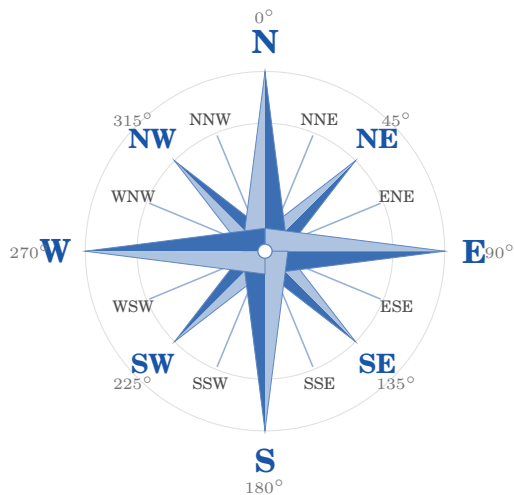
Read off the two new numbers

What is the new **Length**? What is the **Heading**?

This length is **not** the one My Maps gave you. That was **distance**; this is **displacement**.



Part 2 — turning a heading into a direction



Google Earth gives you a **bearing in degrees**, measured **clockwise from north**. It never gives you a letter, so you convert it yourself using the rose.

Round to the nearest whole degree. 312.66° claims far more precision than the measurement deserves — and a bearing landing between two named points takes the in-between name, so 20° is NNE and 110° is ESE.

Worked example

Crail Harbour to school measured 13.57 km on a heading of 312.66° . Written properly, that displacement is 13.6 km **on a bearing of 313°** — or 13.6 km **north-west**.

Now go the other way

Measured *school to Crail* instead, the same line reads 133.54° — south-east. The length does not budge, but the bearing turns through exactly 180° .

Reverse a journey and you reverse its displacement. The distance is unchanged.